Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 20)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 1 . 1 Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 1 . 2 Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

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0 1 . 3 Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 1 . 4 Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

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0 2 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 2 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 2 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 2 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 3 The kinetic energy available to a wind turbine is proportional to the cube of the wind velocity. Table 1 shows mean wind velocities at four possible wind farm sites.

Table 1

Site	Mean wind velocity / m s^{-1}	Distance to grid / km	Other features
A: upland ridge	9	12	Migrating bird route; peat bog
B: coastal farmland	8	3	Near village
C: offshore shallow sea	10	15	Sandbanks
D: inland plain	6	2	Arable farmland

0 3 . 1 Calculate how many times more kinetic energy is available at site C than at site D.

[2 marks]

Answer _____ times

0 3 . 2 Use Table 1 to evaluate the four sites for a large wind farm.

[5 marks]

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0 3 . 3 Give **two** advantages of vertical axis wind turbines (VAWTs) over horizontal axis turbines (HAWTs).

[2 marks]

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0 3 . 4 Explain how direct-drive turbines improve on turbines with a gearbox.

[1 mark]

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0 4 Table 1 lists some techniques used in ecological monitoring.

0 4 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

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0 4 . 2 Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

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0 4 . 3 Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

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0 5 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 5 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 5 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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0 6 Table 1 shows the rate of heat loss through 1 m^2 of different glazing systems for a 1 °C temperature difference (U-value).

Table 1

Glazing	**Heat loss / W m^{-2} K^{-1}**
Single glazing	5.8
Double glazing, air filled	2.8
Double glazing, argon filled, low-e glass	1.6
Triple glazing, krypton filled, low-e glass	0.8
Double glazing, vacuum (0.2 mm gap)	0.74

0 6 . 1 A house has 20 m^2 of single glazing. The average temperature difference between inside and outside over the heating season is 12 °C. Calculate the reduction in the rate of heat loss if the windows are replaced with argon-filled low-e double glazing.

[3 marks]

Answer ..... W

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0 6 . 2 Explain why the gap between the panes should be as large as possible but not too large.

[2 marks]

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0 6 . 3 Explain how low-emissivity glass and inert gas filling each reduce heat loss.

[2 marks]

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0 6 . 4 Explain why vacuum double glazing has a very narrow gap and where it is used.

[2 marks]

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0 6 . 5 State **one** reason why insulation thickness has an optimum value.

[1 mark]

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0 7 . 1 Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

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0 7 . 2 Explain how an oil spill at sea can kill seabirds.

[3 marks]

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0 7 . 3 Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

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0 7 . 4 Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

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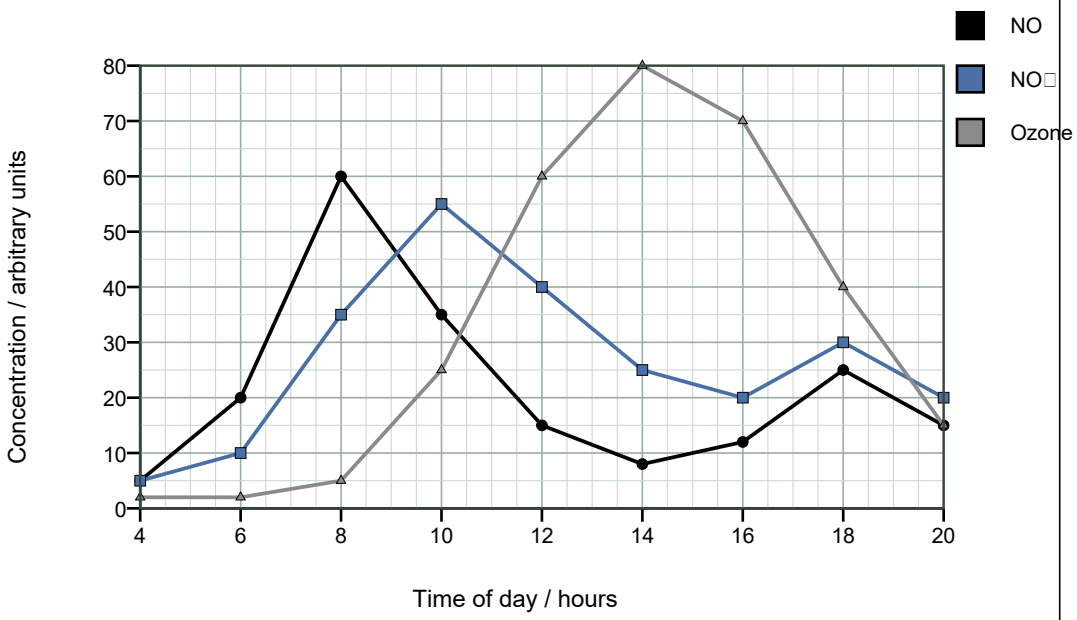
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0 8 Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1



0 8 . 1 Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

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0 8 . 2 Explain why the early-morning peak in NO is followed by a peak in NO_2.

[2 marks]

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0 8 . 3 Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.

[2 marks]

0 8 . 4 Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

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0 9 In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 9 . 1 Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

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0 9 . 2 Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

0 9 . 3 Compare the toxicity of the three chemical forms of mercury.

[3 marks]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Write an essay on one of the following topics.

11 . 1 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

11 . 2 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS